

Euclid, Book I, Proposition I.9

To Bisect a Given Rectilinear Angle

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

1 Introduction

Proposition I.9 presents Euclid’s construction of the bisector of a given rectilinear angle.

The classical proof combines several earlier propositions. A segment of one side of the angle is first copied onto the other side, after which an auxiliary equilateral triangle is constructed. The Side–Side–Side criterion established in Proposition I.8 is then used to prove that the joining segment bisects the angle.

The Hilbert reconstruction follows a different route. Rather than constructing an auxiliary equilateral triangle, it first lays off a segment congruent to one side of the angle, constructs the midpoint of the resulting segment, and finally applies the general Side–Side–Side theorem.

Although the geometric constructions differ substantially, both approaches establish the existence of an angle bisector within neutral geometry.

2 The Classical Configuration

Let $\angle BAC$ be the given rectilinear angle; see Figure 1.

The objective is to construct a ray issuing from the vertex A that divides the angle into two congruent angles.

No assumption is made on the lengths of the sides forming the angle; only the angle itself is given.

2.1 Construction

Choose an arbitrary point D on the side AB .

On the side AC , construct a point E such that

$$AE \cong AD.$$

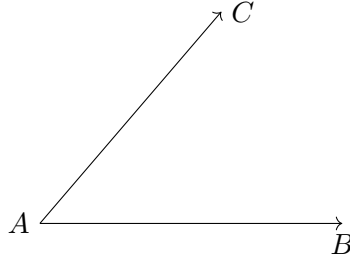


Figure 1: The given angle $\angle BAC$.

Join the points D and E , and construct the equilateral triangle DEF on the segment DE . Finally, join the vertex A to the vertex F ; the completed construction is shown in Figure 2. The claim is that the segment AF bisects the given angle.

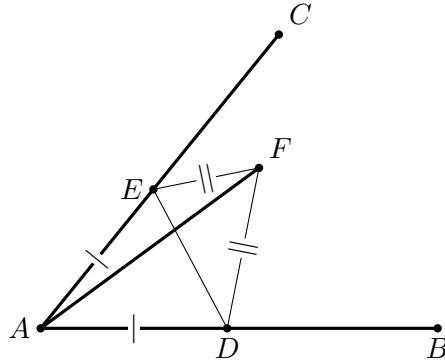


Figure 2: Euclid's construction for Proposition I.9. The points D, E satisfy $AD \cong AE$, and DEF is equilateral. With AF common, SSS for ADF and AEF proves that AF bisects the given angle.

2.2 Application of Proposition I.8

Since the triangle DEF is equilateral,

$$DF \cong EF.$$

Moreover,

$$AD \cong AE$$

by construction, while

$$AF \cong AF$$

is common to the two triangles.

Therefore the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Since D lies on the ray AB and E lies on the ray AC , these are precisely the two parts of the original angle $\angle BAC$.

Hence AF bisects the given angle.

3 Statement

Theorem 1 (Euclid I.9). *Let $\angle BAC$ be a nondegenerate angle.*

Then there exists a point M such that

$$\angle BAM \cong \angle MAC.$$

Equivalently, the ray AM bisects the angle $\angle BAC$.

Diagrammatic audit. Two classical figures are sufficient: the given angle and the completed auxiliary construction. Wilkins's essential diagram is the latter: points D, E are equidistant from A , DEF is equilateral, and AF is the candidate bisector. These are exactly the data used by SSS. No additional classical snapshots are needed. The Hilbert proof, however, uses a genuinely different configuration based on a midpoint of DC ; that difference is recorded in the architecture diagram rather than hidden behind a copy of Euclid's picture.

4 Classical Proof

Choose a point D on the side AB and construct a point E on the side AC such that

$$AD \cong AE.$$

Join the points D and E , and construct the equilateral triangle DEF .

Finally, join the vertex A to the vertex F .

Since

$$AD \cong AE,$$

$$DF \cong EF,$$

and

$$AF \cong AF,$$

the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Because D lies on the ray AB and E lies on the ray AC , these angles are exactly

$$\angle BAF \quad \text{and} \quad \angle FAC.$$

Hence the ray AF bisects the given angle.

This completes the construction.

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces Euclid's auxiliary equilateral triangle by two fundamental constructions already available in the Hilbert interface.

First, a segment congruent to one side of the angle is laid off on the other side of the angle. The midpoint of the resulting segment is then constructed, and the Side-Side-Side theorem is applied to two auxiliary triangles.

Thus the existence of the angle bisector follows entirely from previously established neutral-geometric results.

```

Euclid
-----
angle BAC
|
I.3: choose D,E with AD ~= AE
|
I.1: construct equilateral DEF
|
I.8: SSS on ADF and AEF
|
AF bisects angle BAC

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Hilbert / Lean
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noncollinear A,B,C
|
segment_construction: D on ray AB, AD ~= AC
|
HilbertMidpointExists: M midpoint of DC
|
HilbertSSS: ADM ~= ACM
|
same-ray angle transport: DAM = BAM
|
AM bisects angle BAC

```

6 Hilbert Reconstruction

The formal reconstruction uses a different symmetric configuration from Euclid's. From the noncollinear points A, B, C , segment construction produces a point D on the ray AB such that

$$AD \cong AC.$$

The midpoint theorem is then applied to DC , producing M with

$$D - M - C, \quad DM \cong MC.$$

The triangles ADM and ACM have the three corresponding sides

$$AD \cong AC, \quad DM \cong CM, \quad AM \cong AM.$$

Hilbert SSS therefore yields

$$\angle DAM \cong \angle CAM.$$

Finally, because D lies on the same ray from A as B , angle transport identifies $\angle DAM$ with $\angle BAM$. Hence

$$\angle BAM \cong \angle CAM,$$

which is the required angle bisection.

This route uses neither Euclid's equilateral triangle DEF nor a direct invocation of Proposition I.8. Its symmetry is created instead by segment layoff and midpoint construction.

7 Lean Formalization

The complete implementation is available in the repository. The purpose of the present section is therefore not to reproduce the whole source file, but to isolate the steps that explain the formal geometry.

The first characteristic move is the replacement of Euclid’s pair of equal segments $AD = AE$ by a single laid-off segment on the ray AB :

```
rcases
  HilbertCongruence.segment_construction
    (Geo := Geo)
    A C
    A B
    hAB with
  <D, hRayBD, hAD_AC>
```

Thus D lies on the ray AB and satisfies

$$AD \cong AC.$$

This is the first point at which the Hilbert reconstruction visibly departs from Euclid’s construction.

The second decisive step is the use of a midpoint of DC :

```
rcases
  HilbertMidpointExists
    Geo D C hDC with
  <M, hMid>
```

The midpoint provides

$$D - M - C, \quad DM \cong MC.$$

Together with $AD \cong AC$ and the common side AM , this creates the two triangles used for SSS.

The proof then applies

```
have hSSS :=
  HilbertSSS
    Geo
    A D M
    A C M
    hADM
    hACM
    hAD_AC
    hDM_CM
    hAM_AM
```

and extracts the equality of the angles at A . A final same-ray transport replaces the ray AD by the original ray AB , yielding

$$\angle BAM \cong \angle CAM.$$

The full theorem contains the incidence and noncollinearity bookkeeping needed to justify these steps. Those details remain in the source code; the three excerpts above are the parts that explain the mathematical structure of the Lean proof.

8 Relationship with Earlier Propositions

8.1 Proposition I.1

Euclid begins the construction by choosing points on the two arms of the given angle and constructing an equilateral triangle on the segment joining them. Proposition I.1 supplies the symmetric auxiliary configuration from which the angle bisector will be obtained.

8.2 Proposition I.3

The classical proof uses segment cutting to arrange equal auxiliary lengths on the two sides of the given angle. Proposition I.3 supplies this segment-layoff step.

8.3 Proposition I.8

Proposition I.8 is the decisive congruence theorem in Euclid's proof. The two triangles adjacent to the candidate bisector have three corresponding sides congruent, so SSS yields congruence of the two angles at the original vertex.

Thus the classical dependency pattern is

$$\text{I.3} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{I.9}.$$

8.4 Transition to Proposition I.10

Proposition I.10 uses the angle-bisection construction immediately: an equilateral triangle is erected on the given segment and its apex angle is bisected. SAS then shows that the point where the bisector meets the base is the midpoint.

Thus I.9 becomes constructional infrastructure as soon as it is proved.

9 Hilbert and Book Zero Components Used

9.1 Angle-Bisector Representation

The conclusion is expressed synthetically by a ray from the vertex whose two adjacent angles are congruent. No numerical angle measure is used.

9.2 Segment Layoff

The Hilbert segment-construction machinery supplies the equal auxiliary segments needed to create the symmetric configuration around the given angle.

9.3 SSS

The congruence of the two auxiliary triangles is obtained from the established Hilbert SSS theorem. The desired angle congruence is then a corresponding-angle consequence.

9.4 Interior and Ray Information

The constructed bisector must lie in the intended interior region of the given angle. The formal proof therefore carries ray and side-of-line information that is implicit in Euclid’s diagram.

10 Formalization Notes

Proposition I.9 is the first construction in Book I whose Hilbert reconstruction differs substantially from Euclid’s original proof.

Euclid constructs an auxiliary equilateral triangle and applies the Side–Side–Side criterion established in Proposition I.8. The proof is therefore closely tied to the historical sequence of propositions in the *Elements*.

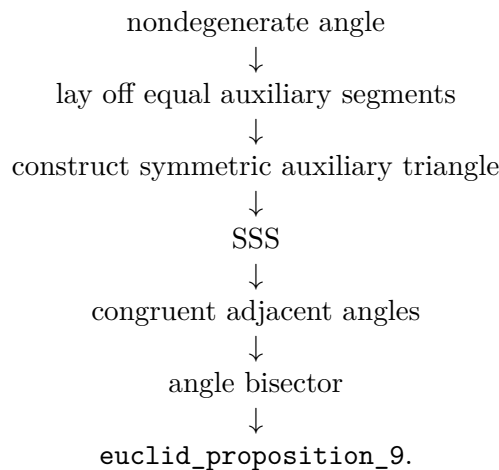
The Hilbert reconstruction adopts a different strategy. A congruent segment is first laid off on one side of the angle, the midpoint of the resulting segment is constructed, and the general Side–Side–Side theorem is applied directly to two auxiliary triangles.

This approach illustrates a general principle of the CGJteamLab library: classical geometric constructions are not reproduced verbatim when a simpler synthetic argument is available within the existing Hilbert theory.

Proposition I.9 therefore demonstrates how a modern axiomatic development can replace a classical construction by a shorter proof while preserving the same mathematical content.

11 Dependency Summary

The formal reconstruction follows the same basic synthetic geometry as the classical proof:



The reconstruction uses only the established Hilbert and Book Zero infrastructure. It introduces

neither numerical angle measure nor a new geometric axiom.

12 Conclusion

Proposition I.9 constructs the bisector of a given rectilinear angle.

The classical proof is an elegant combination of the early construction theorems and SSS: equal auxiliary segments create a symmetric configuration, and triangle congruence converts that symmetry into equality of the two angles at the vertex.

The formal reconstruction preserves this synthetic character while making the positional content explicit. A bisector is not merely an abstract equality of two angle magnitudes; it is represented by a specific ray lying in the appropriate geometric configuration.

I.9 is also an important infrastructural milestone. The construction is used immediately in Proposition I.10 and becomes part of the reusable angle-construction language needed throughout the later development.